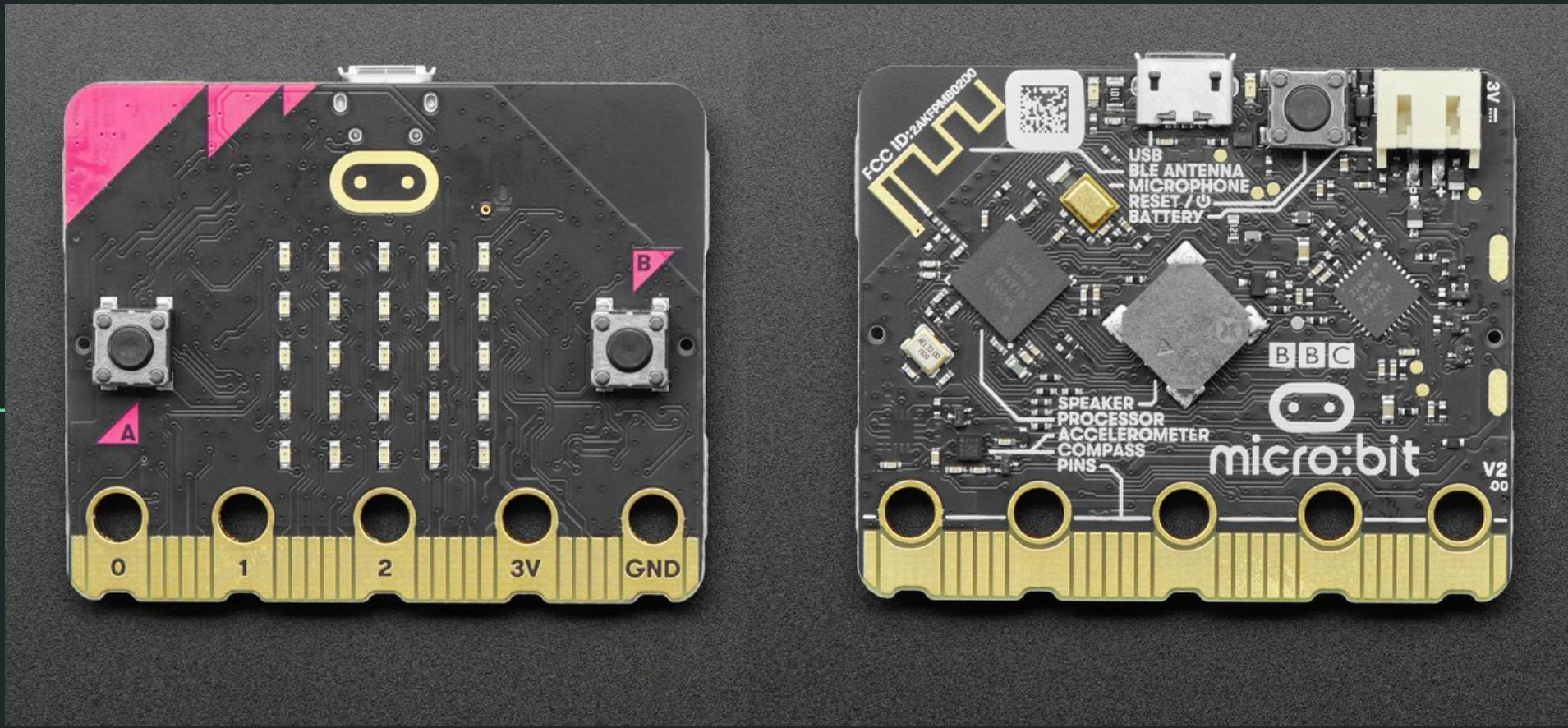
IEEE STEM Learning
BBC micro:bit v3

Steve Case

BBC micro:bit v2

BBC Micro:bit v2 Development Board



BBC micro:bit v1

BBC micro:bit

Introduction

Set up

LEDs and buttons

- Set 1: Icons and animals
- Set 2: Emotions badge
- Set 3: Sunshine

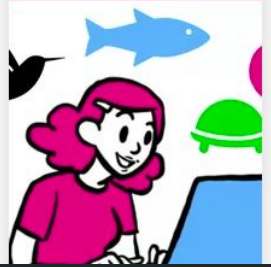
Sensors

Radio and pins

Get creative

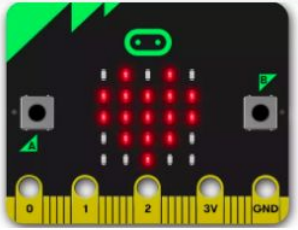
micro:bit

Home learning activities



Set 1: Icons and animals

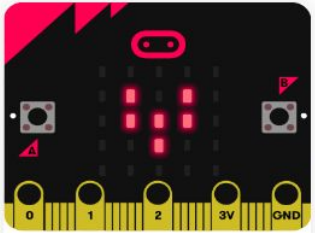
By following this sequence of projects, you'll learn how to create different images on the micro:bit LEDs by sequencing instructions and using the buttons. You'll then bring your creations to life using animation and loops.



Heart

Light up your micro:bit with love by showing a heart

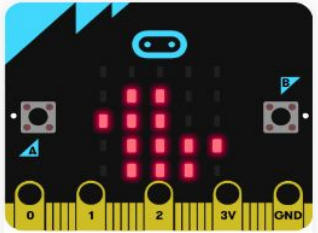
Beginner



Beating heart

Make your micro:bit's heart beat using loops

Beginner



Animated animals

Animate your own animals on the micro:bit display

Beginner

Set 2: Emotions badge

Follow this sequence of projects to create an emotion badge using the LEDs, buttons and accelerometer to let others know how you are feeling. First, you'll program your micro:bit to show happy and sad faces before making them flash and then showing more emotions when your micro:bit is shaken.

BBC Micro:bit

Set up your classroom

Begin your classroom set-up below. Once you've launched your classroom session you will have the option to add your own code to the classroom editor to share with your students.

Name your activity

Give your new activity a name...

Choose a programming language

MakeCode Python

Select storage setting

Use temporary local storage

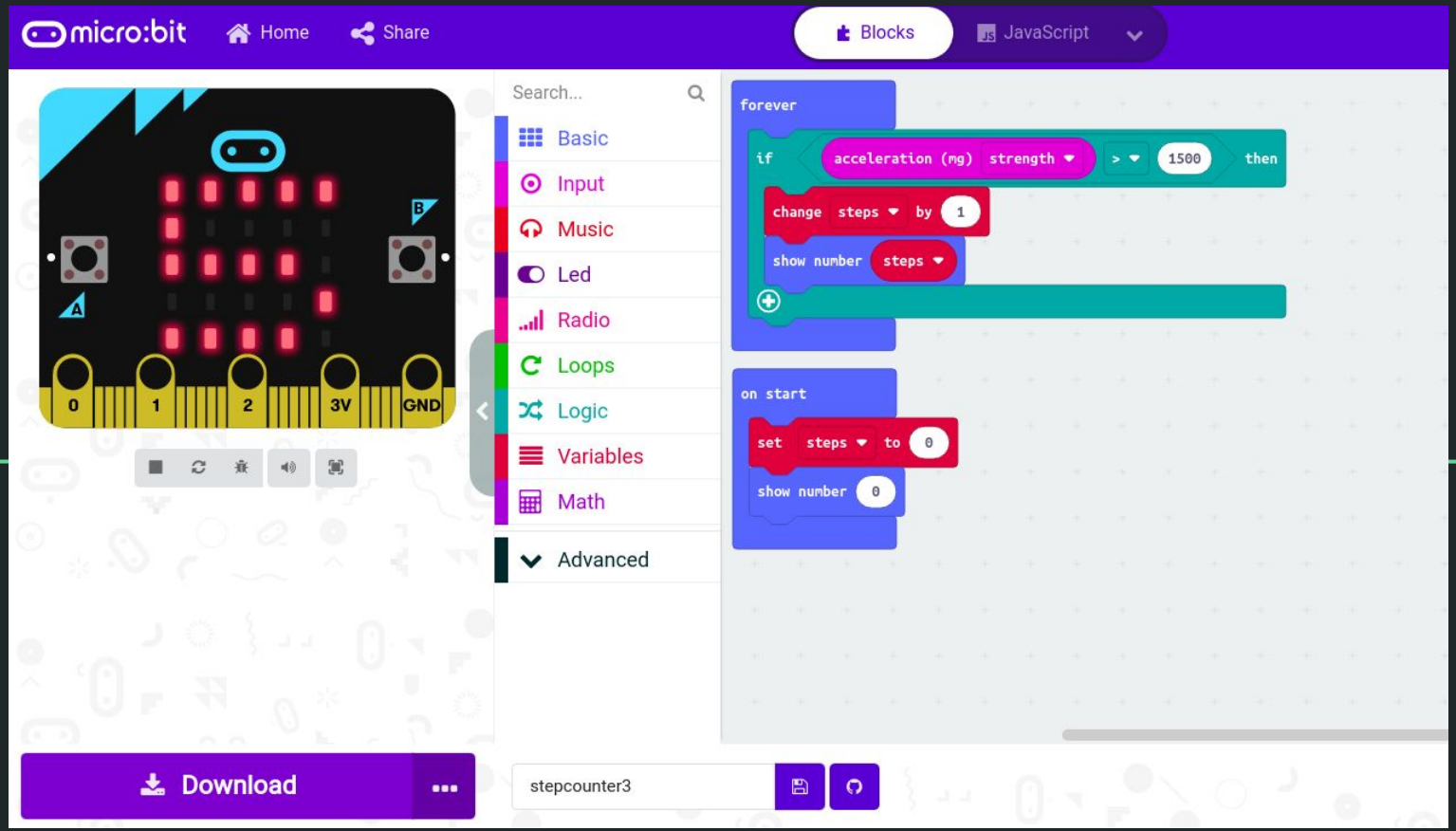
Launch classroom

Explore other activities

Want to resume an activity?

BBC micro:bit

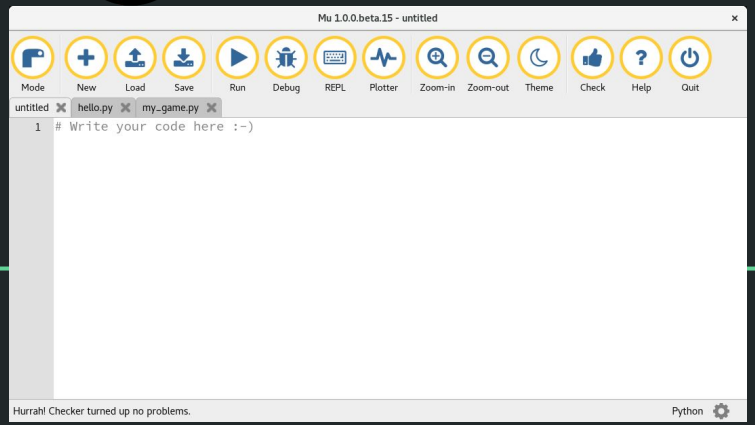
Microsoft MakeCode



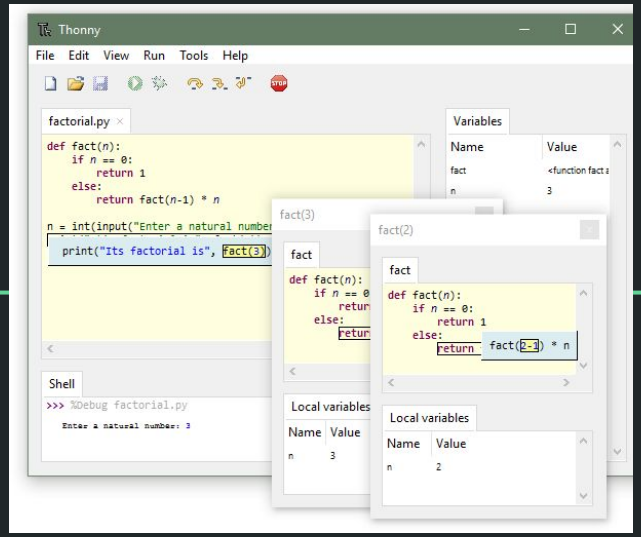
BBC Micro:bit

Python Editors

Mu Editor



Thonny



Python Editor Thonny

File Edit View Run Tools Help

Program tabs

Files

This computer
/home/steve/microbit/
students

- robot.py
- sound2.py
- speech.py
- temperature1.py
- temperature2.py
- weather_station.py
- while_loop1.py

micro:bit

- 8ball_1.py
- disp_n.py
- eight_ball.py
- for_loop1.py
- main.py

dice.py [main.py]

```
1  #
2  #  For loop demo with display
3  #
4
5  from microbit import *
6  import random
7  display.clear()
8
9
10 while True:
11     for i in range(0, 5):
```

Shell Exception Program tree TODO

```
>>> %Run -c $EDITOR_CONTENT

MPY: soft reboot
Traceback (most recent call last):
  File "<stdin>", line 20, in <module>
KeyboardInterrupt:
>>>
```

REPL

MicroPython (BBC micro:bit) • BBC micro:bit CMSIS-DAP @ /dev/ttyACM0

Images - import microbit, dir(Image)

'ALL_ARROWS', 'ALL_CLOCKS', 'ANGRY', 'ARROW_E', 'ARROW_N', 'ARROW_NE', 'ARROW_NW', 'ARROW_S', 'ARROW_SE', 'ARROW_SW', 'ARROW_W', 'ASLEEP', 'BUTTERFLY', 'CHESSBOARD', 'CLOCK1', 'CLOCK10', 'CLOCK11', 'CLOCK12', ' CLOCK2', 'CLOCK3', 'CLOCK4', CLOCK5', 'CLOCK6', CLOCK7', 'CLOCK8', CLOCK9', 'CONFUSED', 'COW', 'DIAMOND', 'DIAMOND_SMALL', 'DUCK', 'FABULOUS', 'GHOST', 'GIRAFFE', 'HAPPY', 'HEART', 'HEART_SMALL', 'HOUSE', 'MEH', 'MUSIC_CROTCHET', 'MUSIC_QUAVER', 'MUSIC_QUAVERS', 'NO', 'PACMAN', 'PITCHFORK', 'RABBIT', 'ROLLERSKATE', 'SAD', 'SCISSORS', 'SILLY', 'SKULL', 'SMILE', 'SNAKE', 'SQUARE','SQUARE_SMALL" STICKFIGURE', 'SURPRISED', 'SWORD', 'TARGET', 'TORTOISE', 'TRIANGLE', 'TRIANGLE_LEFT', 'TSHIRT', 'UMBRELLA', 'XMAS', 'YES',

Music - import music, dir(music)

'BADDY', 'BA_DING',
'BIRTHDAY', 'BLUES',
'CHASE', 'DADADADUM',
'ENTERTAINER',
'FUNERAL', 'FUNK',
'JUMP_DOWN', 'JUMP_UP',
'NYAN', 'ODE',
'POWER_DOWN',
'POWER_UP', 'PRELUDE',
'PUNCHLINE', 'PYTHON',
'RINGTONE',
'WAWAWAWAA',
'WEDDING', 'get_tempo

Reference Material

BBC micro:bit www.microbit.org

IEEE Course Material & Programs
https://github.com/IEEE-BV/microbit

Thonny-Editor Installation:
https://thonny.org/

Adafruit Industries: www.adafruit.com
Sparkfun electronics: www.sparkfun.com

Python:
Micro Python documentation for micro:bit
https://microbit-micropython.readthedocs.io/en/v2-docs/#

Other References:
https://www.w3schools.com/python/default.asp
https://www.python.org/ (advanced full reference for Python)
https://wiki.python.org/moin/BeginnersGuide (still pretty advanced)